Chapter 5, Section 6

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August 14, 2026

1. Let X be a surface in $\mathbb{P}^n$, $n \geqslant 3$, defined as the complete intersection of hypersurfaces of degree $d_1, \dots, d_{n-2}$, with each $d_i \geqslant 2$. Show that for all but finitely many choices of $(n, d_1, \dots, d_{n-2})$, the surface X is of general type. List the exceptional cases, and where they fit into the classification picture.

2. Prove the following theorem of Chern and Griffiths. Let X be a nonsingular surface of degree d in $\mathbb{P}^{n+1}_{\mathbb{C}}$, which is not contained in any hyperplane. If $d < 2n$, then $p_g(X) = 0$. If $d = 2n$, then either $p_g(X) = 0$, or $p_g(X) = 1$ and X is a K3 surface.